

JEE MAIN GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)
Time: 3HRS

Date:16-01-24
Max. Marks: 300

Name of the Student: _____

H.T. NO:

16-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-18(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

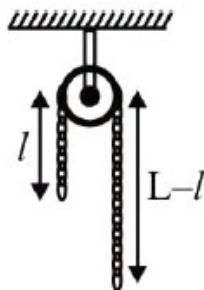
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In an adiabatic process, $R = \frac{2}{3}C_v$. The pressure of the gas will be proportional to:

1) $T^{5/3}$
2) $T^{5/2}$
3) $T^{5/4}$
4) $T^{5/6}$
2. At time $t = 0$ a particle starts travelling from a height $7\hat{z} \text{ cm}$ in a plane keeping z coordinate constant. At any instant of time its position along the x and y directions are defined as $3t$ and $5t^3$ respectively. At $t = 1\text{s}$ acceleration of the particle will be

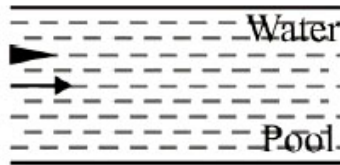
1) $-30\hat{y}$
2) $30\hat{y}$
3) $3\hat{x} + 15\hat{y}$
4) $3\hat{x} + 15\hat{y} + 7\hat{z}$
3. A pressure-pump has a horizontal tube of cross-sectional area 10 cm^2 for the outflow of water at a speed of 20 m/s . The force exerted on the vertical wall just in front of the tube which stops water horizontally flowing out of the tube, is: [given : density of water = 1000 kg/m^3]

1) 300 N
2) 500 N
3) 250 N
4) 400 N
4. A uniform metal chain of mass m and length ' L ' passes over a massless and frictionless pulley. It is released from rest with a part of its length ' l ' is hanging on one side and rest of its length ' $L-l$ ' is hanging on the other side of the pulley. At a certain point of time, when $l = \frac{L}{x}$, the acceleration of the chain is $\frac{g}{2}$. The value of x is

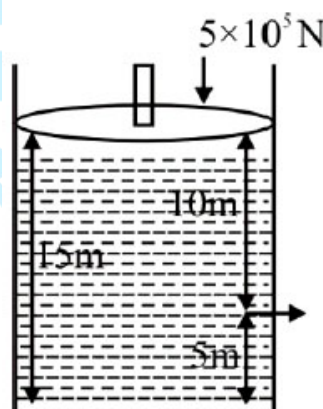


- 1) 6
2) 2
3) 1.5
4) 4

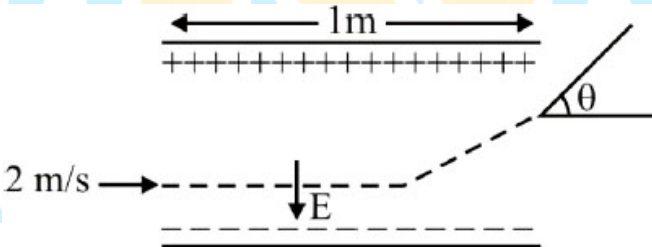
5. A bullet of mass 200 g having initial kinetic energy 90 J is shot inside a long swimming pool providing uniform retardation as shown in the figure. If its kinetic energy reduces to 40 J within 1s, the minimum length of the pool, the bullet has to travel so that it completely comes to rest is



- 1) 45 m 2) 90 m 3) 125 m 4) 25 m
6. Assume there are two identical simple pendulum Clocks-1 is placed on the earth and Clock-2 is placed on a space station located at a height h above the earth surface. Clock-1 and Clock-2 operate at time periods 4s and 6s respectively. Then the value of h is (consider radius of earth $R_E = 6400$ km and g on earth $= 10 \text{ m/s}^2$)
- 1) 1200 km 2) 1600 km 3) 3200 km 4) 4800 km
7. Consider a cylindrical tank of radius 1m is filled with water. The top surface of water is at 15 m from the bottom of the cylinder. There is a hole on the wall of cylinder at a height of 5m from the bottom. A force of $5 \times 10^5 \text{ N}$ is applied on the top surface of water using a piston. The speed of efflux from the hole will be :
(given atmospheric pressure $P_A = 1.01 \times 10^5 \text{ Pa}$, density of water $\rho_w = 1000 \text{ kg/m}^3$ and gravitational acceleration $g = 10 \text{ m/s}^2$)

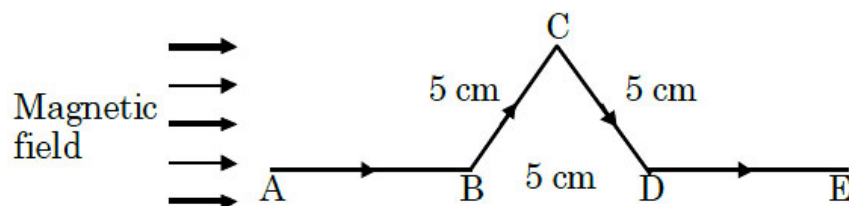


- 1) 11.6 m/s 2) 10.8 m/s 3) 17.8 m/s 4) 14.4 m/s

8. A vessel contains 14 g of nitrogen gas at a temperature of 27°C . The amount of heat to be transferred to the gas to double the r.m.s. speed of its molecules will be :
(Take $R = 8.32 \text{ J mol}^{-1}\text{K}^{-1}$)
- 1) 2229 J 2) 5616 J 3) 9360 J 4) 13,104 J
9. A slab of dielectric constant K has the same cross-sectional area as the plates of a parallel plate capacitor and thickness $\frac{3}{4}d$, where d is the separation of the plates. The capacitance of the capacitor when the slab is inserted between the plates will be :
(Given C_0 = capacitance of capacitor with air as medium between plates.)
- 1) $\frac{4KC_0}{3+K}$ 2) $\frac{3KC_0}{3+K}$ 3) $\frac{3+K}{4KC_0}$ 4) $\frac{K}{4+K}$
10. A uniform electric field $E = (8m/e) \text{ V/m}$ is created between two parallel plates of length 1m as shown in figure, (where m = mass of electron and e = charge of electron). An electron enters the field symmetrically between the plates with a speed of $2m/s$. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be.....
- 
- 1) $\tan^{-1}(4)$ 2) $\tan^{-1}(2)$ 3) $\tan^{-1}\left(\frac{1}{3}\right)$ 4) $\tan^{-1}(3)$
11. Given below are two statements :
- Statement I :** A uniform wire of resistance 80Ω is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be 5Ω .
- Statement II :** Two resistance $2R$ and $3R$ are connected in parallel in a electric circuit. The value of thermal energy developed in $3R$ and $2R$ will be in the ratio $3 : 2$.
- In the light of the above statements, choose the most appropriate answer from the options given below

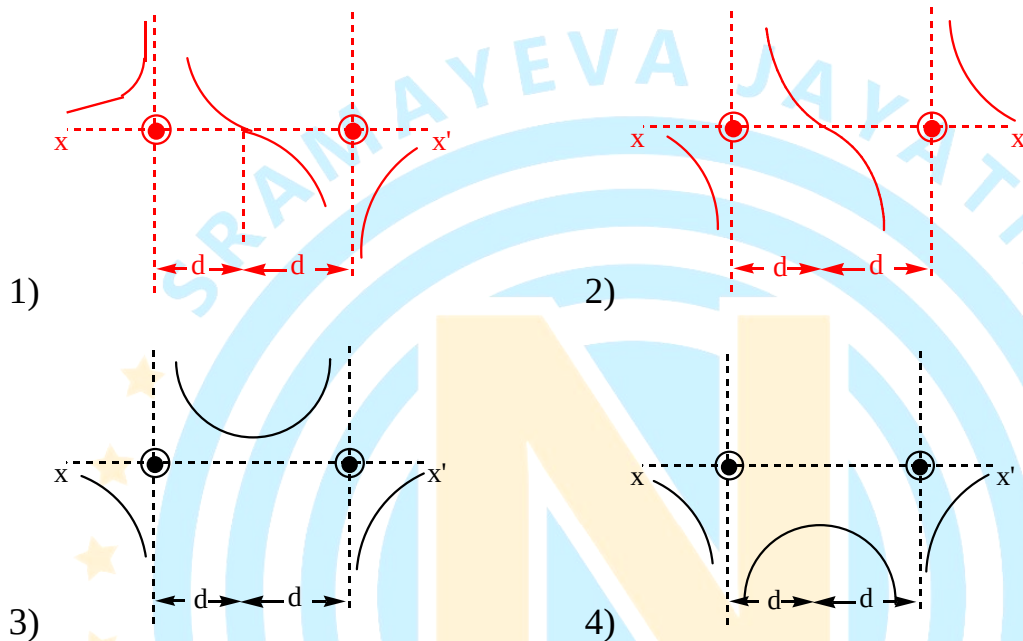
- 1) Both statement I and statement II are correct
- 2) Both statement I and statement II are incorrect
- 3) Statement I is correct but statement II is incorrect
- 4) Statement I is incorrect but statement II is correct.

12. A triangular shaped wire carrying 10A current is placed in a uniform magnetic field of 0.5T, as shown in figure. The magnetic force on segment CD is
(Given $BC = CD = BD = 5$ cm).

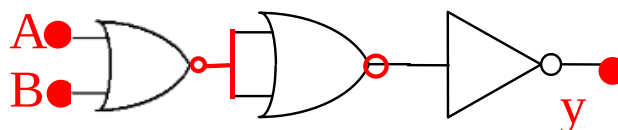


- 1) 0.126 N 2) 0.312 N 3) 0.216 N 4) 0.245 N
13. The magnetic field at the center of current carrying circular loop is B_1 . The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2 . The value of B_1 / B_2 will be
- 1) 9 : 4 2) $12 : \sqrt{5}$ 3) 8 : 1 4) $5 : \sqrt{3}$
14. The charges appearing on inner surfaces of a parallel plate capacitor are q_1 and q_2 . The charges appearing on outer surfaces of the capacitor are equal. Find the potential difference between plates of the capacitor, if capacitance is C.
- 1) $\frac{q_1 + q_2}{2C}$ 2) $\frac{q_1 - q_2}{2C}$ 3) $\frac{q_1 + q_2}{C}$ 4) $\frac{q_1 - q_2}{C}$
15. Sun light falls normally on a surface of area 36 cm^2 and exerts an average force of $7.2 \times 10^{-9} \text{ N}$ within a time period of 20 minutes. Considering a case of complete absorption, the energy flux of incident light is
- 1) $25.95 \times 10^2 \text{ W / cm}^2$ 2) $8.64 \times 10^{-6} \text{ W / cm}^2$ 3) 6.0 W / cm^2 4) 0.06 W / cm^2
16. The power of a lens (biconvex) is 1.25 m^{-1} in particular medium. Refractive index of the lens is 1.5 and radii of curvature are 20 cm and 40 cm respectively. The refractive index of surrounding medium :
- 1) 1.0 2) $\frac{9}{7}$ 3) $\frac{3}{2}$ 4) $\frac{4}{3}$

17. Two streams of photons, possessing energies five and ten times the work function of metal are incident on the metal surface successively. The ratio of the maximum velocities of the photoelectron emitted, in the two cases respectively, will be
- 1) 1 : 2 2) 1 : 3 3) 2 : 3 4) 3 : 2
18. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of magnetic field B along the line XX' is given by:



19. When 100 V dc is applied across a coil, a current of 1A flows through it. When 100V ac of 50 Hz is applied across the same coil, only 0.5 A flows. The resistance and inductance of the coil are (take $\pi^2 = 10$)
- 1) $50\Omega, 0.3H$ 2) $50\Omega, \sqrt{(0.3)}H$
- 3) $100\Omega, 0.3H$ 4) $100\Omega, \sqrt{(0.3)}H$
20. The given network is equivalent to:



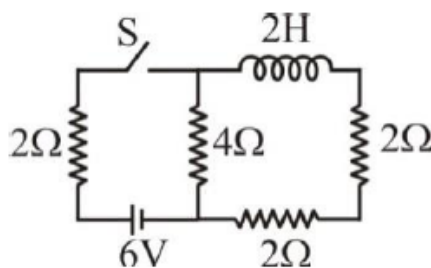
- 1) OR gate 2) NOR gate 3) NOT gate 4) AND gate

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

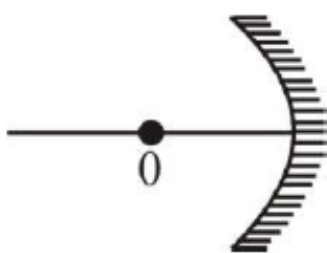
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

21. A ball is thrown vertically upwards with a velocity of 19.6 ms^{-1} from the top of a tower. The ball strikes the ground after 6 s. The height from the ground up to which the ball can rise will be $\left(\frac{k}{5}\right)m$.
The value of k is (use $g = 9.8 \text{ m/s}^2$)
22. The distance of centre of mass from end A of a one dimensional rod (AB) having mass density $\rho = \rho_0 \left(1 - \frac{x^2}{L^2}\right) \text{ kg/m}$ and length L (in meter) is $\frac{3L}{\alpha}m$. The value of α is
(where x is the distance from end A)
23. A string of area of cross-section 4 mm^2 and length 0.5 is connected with a rigid body of mass 2 kg. The body is rotated in a vertical circular path of radius 0.5 m. The body acquires a speed of 5 m/s at the bottom of the circular path. Strain produced in the string when the body is at the bottom of the circle is $\times 10^{-5}$. (Use Young's modulus 10^{11} N/m^2 and $g = 10 \text{ m/s}^2$)
24. At a certain temperature, the degrees of freedom per molecule for gas is 8. The gas performs 150 J of work when it expands under constant pressure. The amount of heat absorbed by the gas will be J.
25. The potential energy of a particle of mass 4 kg in motion along the x-axis is given by $U = 4(1 - \cos 4x) \text{ J}$. The time period of the particle for small oscillation ($\sin \theta \approx \theta$) is $\left(\frac{\pi}{K}\right)s$.
The value of K is
26. An electrical bulb rated 220 V, 100 W, is connected in series with another bulb rated 220 V, 60 W. If the voltage across combination is 220 V, the power consumed by the 100 W bulb will be about W.
27. For the given circuit the current through battery of 6 V just after closing the switch 'S' will be A.



28. An object 'o' is placed at a distance of 100 cm in front of a concave mirror of radius of curvature 200 cm as shown in the figure. The object starts moving towards the mirror at a speed 2 cm/s. The position of the image from the mirror after 10s will be at cm.



29. In an experiment with a convex lens, the plot of the image distance (v) against the object distance (u) measured from the focus gives a curve $uv = 225$. If all the distances are measured in cm. The magnitude of the focal length of the lens is cm.
30. In an experiment to find acceleration due to gravity (g) using simple pendulum, time period of 0.5 s is measured from time of 100 oscillation with a watch of 1s resolution. If measured value of length is 10 cm known to 1mm accuracy. The accuracy in the determination of g is found to be $x\%$. The value of x is

THE NARAYANA GROUP

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R

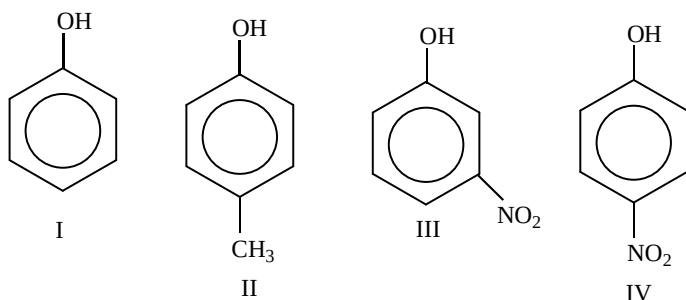
Assertion A : Zero orbital overlap is an out of phase overlap.

Reason : It results due to different orientation/direction of approach of orbitals.

In the light of the above statements. Choose the correct answer from the options given below

- 1) Both A and R are true and R is the correct explanation of A
 - 2) Both A and R are true but R is NOT the correct explanation of A
 - 3) A is true but R is false
 - 4) A is false but R is true
32. The correct decreasing order for metallic character is
- 1) $\text{Na} > \text{Mg} > \text{Be} > \text{Si} > \text{P}$
 - 2) $\text{P} > \text{Si} > \text{Be} > \text{Mg} > \text{Na}$
 - 3) $\text{Si} > \text{P} > \text{Be} > \text{Na} > \text{Mg}$
 - 4) $\text{Be} > \text{Na} > \text{Mg} > \text{Si} > \text{P}$
33. Dipole moment of HCl molecule is found to be 0.816D. Assuming HCl bond length to be equal to $1\text{\AA}$, the % ionic character of HCl molecule is:
- 1) 10%
 - 2) 17%
 - 3) 27%
 - 4) 37%
34. The degree of dissociation (α) of a weak electrolyte, A_xB_y is related to Vant't Hoff factor (i) by the expression:
- 1) $\alpha = \frac{i-1}{(x+y-1)}$
 - 2) $\alpha = \frac{i-1}{(x+y+1)}$
 - 3) $\alpha = \frac{x+y-1}{i-1}$
 - 4) $\alpha = \frac{x+y+1}{i-1}$

35. In the following compounds, the order of acidity is:

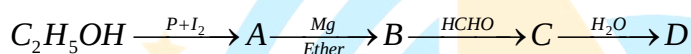


- 1) II>IV>I>II 2) I>IV>III>II 3) II>I>III>IV 4) IV>III>I>II

36. White phosphorus reacts with thionyl chloride to give

- 1) PCl_5 , SO_2 and S_2Cl_2 2) PCl_3 , SO_2 and S_2Cl_2
 3) PCl_3 , SO_2 and Cl_2 4) PCl_5 , SO_2 and Cl_2

37. The product in the reaction is:



- 1) propanal 2) butanal 3) n-butanol 4) n-propanol

38. Which of the following pair is not isoelectronic species?

(At. no. Sm, 62; Er, 68; Yb, 70; Lu, 71; Eu, 63; Tb, 65; Tm, 69)

- 1) Sm^{2+} and Eu^{3+} 2) Yb^{2+} and Lu^{3+} 3) Eu^{2+} and Tb^{4+} 4) Tb^{2+} and Tm^{4+}

39. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Permanganate titrations are not performed in presence of hydrochloric acid.

Reason R : Chlorine is formed as a consequence of oxidation of hydrochloric acid.

In the light of the above statements, choose the correct answer from the options given below

- 1) Both A and R are true and R is the correct explanation of A
 2) Both A and R are true but R is NOT the correct explanation of A
 3) A is true but R is false
 4) A is false but R is true

40. Match List I with List II

	List I (Complex)		List II (Hybridization)
A.	$Ni(CO)_4$	I	sp^3
B.	$[Ni(CN)_4]^{2-}$	II	sp^3d^2
C.	$[Co(CN)_6]^{3-}$	III	d^2sp^3
D.	$[CoF_6]^{3-}$	IV	dsp^2

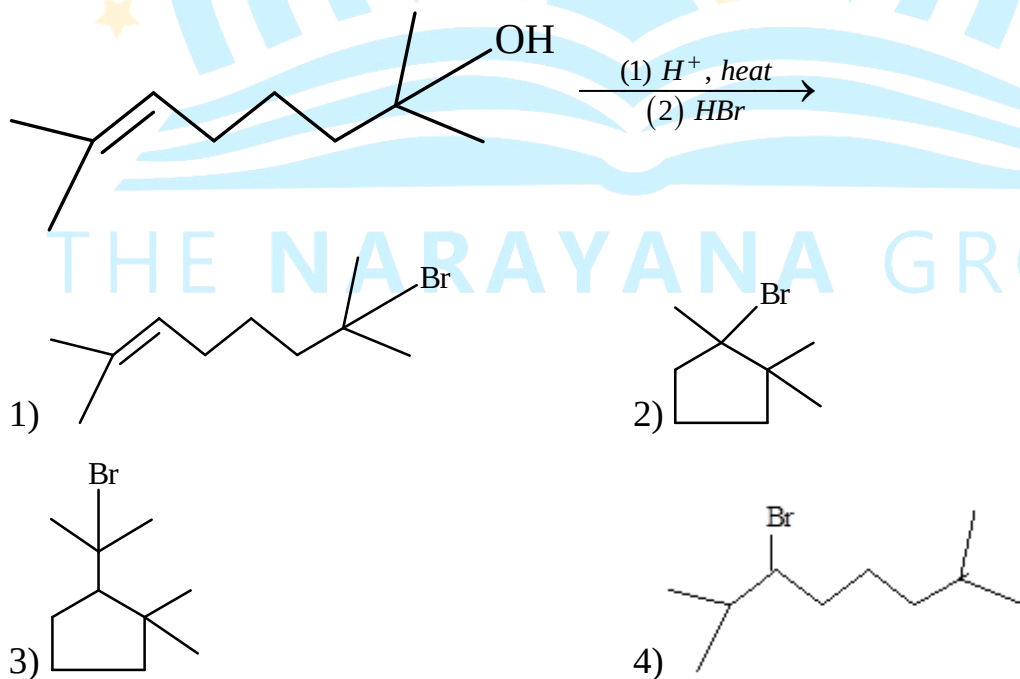
Choose the correct answer from the options given below:

- 1) A-IV, B-I, C-III, D-II 2) A-I, B-IV, C-III, D-II
 3) A-I, B-IV, C-II, D-III 4) A-IV, B-I, C-II, D-III

41. Dinitrogen and dioxygen, the main constituents of air do not react with each other in atmosphere to form oxides of nitrogen because

- 1) N_2 is unreactive in the condition of atmosphere
 2) Oxides of nitrogen are unstable.
 3) Reaction between them can occur in the presence of a catalyst.
 4) The reaction is endothermic and require very high temperature.

42. The major product in the given reaction is



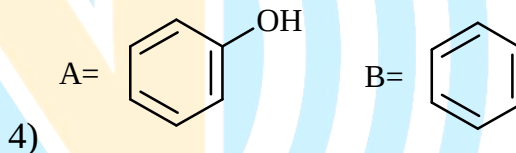
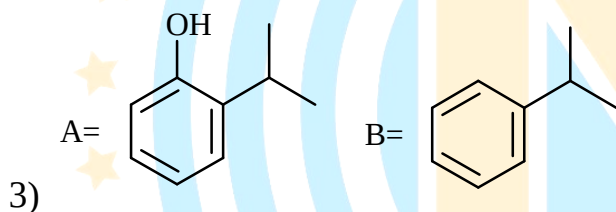
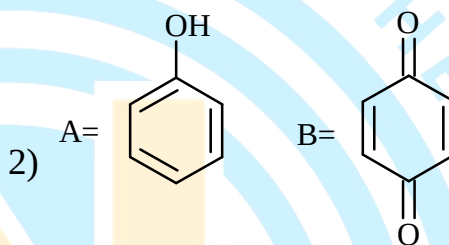
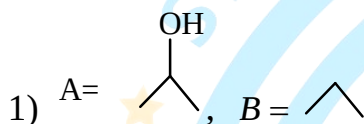
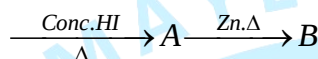
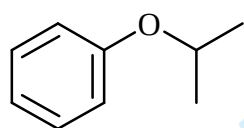
43. Arrange the following in increasing order of reactivity towards nitration

- A. p-xylene B. bromobenzene C. mesitylene D, nitrobenzene
E. benzene

Choose the correct answer from the options given below

- 1) $C < D < E < A < B$ 2) $D < B < E < A < C$
3) $D < C < E < A < B$ 4) $C < D < E < B < A$

44. Compound I is heated with Conc. HI to give a hydroxy compound A which is further heated with Zn dust to give compound B. Identify A and B.



45. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Aniline on nitration yields ortho, meta & para nitro derivatives of aniline.

Reason R: Nitrating mixture is a strong acidic mixture.

In the light of the above statements, choose the correct answer from the options given below

- 1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true but R is NOT the correct explanation of A
3) A is true but R is false
4) A is false but R is true

46. Which one represents an impossible arrangement?

	n	l	m_l	s
1)	3	2	-2	1/2
2)	4	0	0	1/2
3)	3	2	-3	1/2
4)	5	3	0	1/2

47. On increasing temperature, the equilibrium constant of exothermic and endothermic reactions would respectively

- 1) Increases and decreases
- 2) Decreases and increases
- 3) Increases and increases
- 4) Decreases and decrease

48. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R

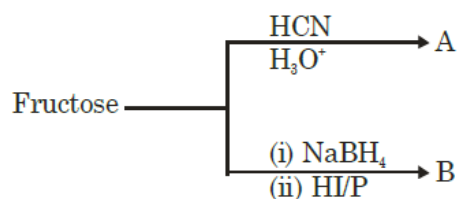
Assertion A : Thin layer chromatography is an adsorption chromatography.

Reason : A thin layer of silica gel is spread over a glass plate of suitable size in thin layer chromatography which acts as an adsorbent.

In the light of the above statements, choose the correct answer from the options given below

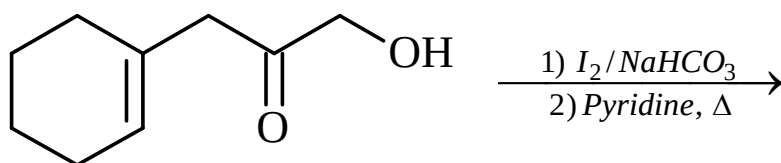
- 1) Both A and R are true and R is the correct explanation of A
- 2) Both A and R are true but R is NOT the correct explanation of A
- 3) A is true but R is false
- 4) A is false but R is true

49. The formulas of A and B for the following reaction sequence are



- 1) $A = C_7H_{14}O_8$, $B = C_6H_{14}$
- 2) $A = C_7H_{13}O_7$, $B = C_7H_{14}O$
- 3) $A = C_7H_{12}O_8$, $B = C_6H_{14}$
- 4) $A = C_7H_{14}O_8$, $B = C_6H_{14}O_6$

50.



Find out the major product for the above reaction

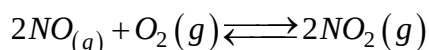


SECTION-II (NUMERICAL VALUE ANSWER TYPE)

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51. 2L of 0.2 M H_2SO_4 is reacted with 2L of 0.1 M NaOH solution, the molarity of the resulting product Na_2SO_4 in the solution is _____ millimolar. (Nearest integer).
52. The oxidation number of Co in $[Co(en)_3]_2(SO_4)_3$ is _____.
53. If the wavelength for an electron emitted from H-atom is $3.3 \times 10^{-10} m$, then energy absorbed by the electron in its ground state compared to minimum energy required for its escape from the atom, is _____ times. (Nearest integer).
[Given : $h = 6.626 \times 10^{-34} Js$, Mass of electron = 9.1×10^{-31}]
54. Maximum number of hydrogen bonds formed by an ammonia molecule in liquid ammonia is _____.
55. At 600K, 2 mol of NO are mixed with 1 mol of O_2 .

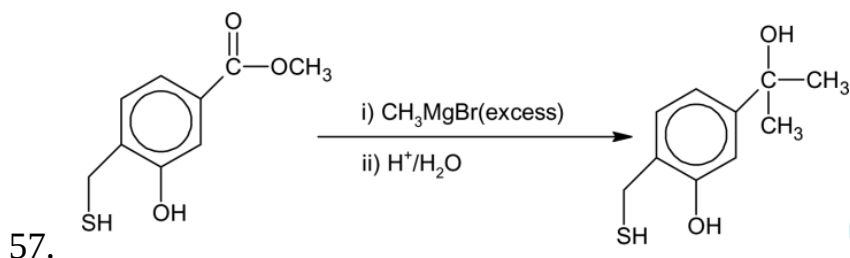


The reaction occurring as above comes to equilibrium under a total pressure of 1 atm. Analysis of the system shows that 0.6 mol of oxygen are present at equilibrium. The equilibrium constant for the reaction is _____. (Nearest integer).

56. A sample of 0.125 g of an organic compound when analysed by Duma's method yields 22.78 mL of nitrogen gas collected over KOH solution at 280K and 759 mm Hg. The percentage of nitrogen in the given organic compound is _____. (Nearest integer).

1) The vapour pressure of water at 280 K is 14.2 mm Hg

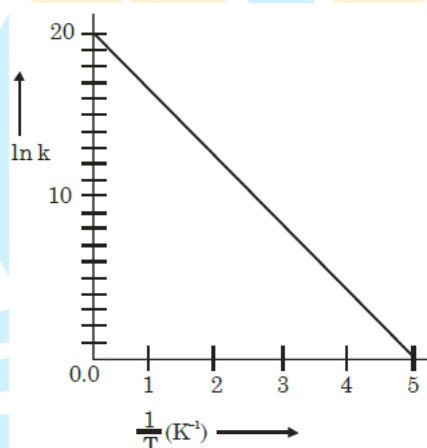
2) $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$



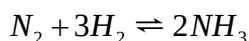
Moles of CH_3MgBr consumed per mole of reactant will be

58. For a reaction, given below is the graph of $\ln k$ vs $\frac{1}{T}$. The activation energy for the reaction is equal to _____ cal mol^{-1} . (Nearest integer).

(Given : $R = 2 \text{ cal K}^{-1} \text{ mol}^{-1}$)



59. 28 g of N_2 and 6 g of H_2 were kept at 400°C in 1 litre vessel, the equilibrium mixture contained 27.54g of NH_3 . The approximate value of K_c for the above reaction is (in $\text{mole}^{-2} \text{ litre}^2$)



60. Among the following the number of state variable is _____.

Internal energy (U)

Volume (V)

Heat (q)

Enthalpy (H)

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(SINGLE CORRECT ANSWER TYPE)

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61. Let $S = \left\{ x \in [-6, 3] - \{-2, 2\} : \frac{|x+3|-1}{|x|-2} \geq 0 \right\}$ and $T = \{x \in \mathbb{Z} : x^2 - 7|x| + 9 \leq 0\}$. Then the number of elements in $S \cap T$ is
- 1) 7 2) 5 3) 4 4) 3
62. If x, y, z are three non zero real numbers satisfying the condition $x+y+z = xyz$, then the value of $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z$ is
- 1) π 2) $\frac{\pi}{2}$ 3) $\frac{3\pi}{4}$ 4) $\frac{3\pi}{2}$
63. Let A and B be any two 3×3 symmetric and skew symmetric matrices respectively. Then which of the following is NOT true?
- 1) $A^4 - B^4$ is a symmetric matrix 2) $AB - BA$ is a symmetric matrix
- 3) $B^5 - A^5$ is a skew-symmetric matrix 4) $AB + BA$ is a skew-symmetric matrix
64. Let $f(x) = ax^2 + bx + c$ be such that $f(1) = 3$, $f(-2) = \lambda$ and $f(3) = 4$. If $f(0) + f(1) + f(-2) + f(3) = 14$, then λ is equal to
- 1) -4 2) $\frac{13}{2}$ 3) $\frac{23}{2}$ 4) 4
65. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \lim_{n \rightarrow \infty} \frac{\cos(2\pi x) - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$ ($n \in \mathbb{N}$) is continuous for all x in
- 1) $\mathbb{R} - \{-1\}$ 2) $\mathbb{R} - \{-1, 1\}$ 3) $\mathbb{R} - \{1\}$ 4) $\mathbb{R} - \{0\}$
66. The function $f(x) = xe^{x(1-x)}$, $x \in \mathbb{R}$, is
- 1) increasing in $\left(-\frac{1}{2}, 1\right)$ 2) decreasing in $\left(\frac{1}{2}, 2\right)$
- 3) increasing in $\left(-1, -\frac{1}{2}\right)$ 4) decreasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$

67. The sum of the absolute maximum and absolute minimum values of the function

$f(x) = \tan^{-1}(\sin x - \cos x)$ in the interval $[0, \pi]$ is

- 1) 0 2) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) - \frac{\pi}{4}$ 3) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right) - \frac{\pi}{4}$ 4) $\frac{-\pi}{12}$

68. $y = \tan^{-1}\left(\frac{\sqrt{1+a^2x^2}-1}{ax}\right) \Rightarrow (1+a^2x^2)y_2 + 2a^2xy_1 =$

(Where y_1, y_2 represents first order, second order derivating) ?

- 1) a^2 2) $2a^2$ 3) 0 4) $-2a^2$

69. $I_n = \int_0^{\pi} \tan^n x \, dx$, Then $I_6 + I_8 =$

- 1) $\frac{1}{7}$ 2) $\frac{2}{7}$ 3) $\frac{1}{6}$ 4) $\frac{1}{8}$

70. The area inside the parabola $5x^2 - y = 0$ but outside the parabola $2x^2 - y + 9 = 0$

- 1) $12\sqrt{3}$ sq units 2) $6\sqrt{3}$ sq. units
3) $8\sqrt{3}$ sq units 4) $4\sqrt{3}$ sq units

71. Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \frac{1}{x^2-1}y = \left(\frac{x-1}{x+1}\right)^{\frac{1}{2}}$, $x > 1$

passing through the point $\left(2, \sqrt{\frac{1}{3}}\right)$. Then $\sqrt{7}y(8)$ is equal to

- 1) $11 + 6\log_e 3$ 2) 19 3) $12 - 2\log_e 3$ 4) $19 - 6\log_e 3$

72. The differential equation of the family of circles passing through the points $(0, 2)$ and

$(0, -2)$ is

- 1) $2xy \frac{dy}{dx} + (x^2 - y^2 + 4) = 0$ 2) $2xy \frac{dy}{dx} + (x^2 + y^2 - 4) = 0$
3) $2xy \frac{dy}{dx} + (y^2 - x^2 + 4) = 0$ 4) $2xy \frac{dy}{dx} - (x^2 - y^2 + 4) = 0$

73. The mean of n items is $\bar{X}$. If first item is increased by 1, second item by 2, third item by 3 and so on, then new mean is

- 1) $\bar{X} + \frac{n+1}{2}$ 2) $\bar{X} + \frac{n}{2}$ 3) $\bar{X} + n$ 4) $\bar{X} + \frac{n-1}{2}$

74. Let the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ pass through the point $(2\sqrt{2}, -2\sqrt{2})$. A parabola is drawn whose focus is same as the focus of H with positive abscissa and the directrix of the parabola passes through the other focus of H . If the length of the latus rectum of the parabola is e times the length of the latus rectum of H , where e is the eccentricity of H , then which of the following points lies on the parabola?
- 1) $(2\sqrt{3}, 3\sqrt{2})$ 2) $(3\sqrt{3}, -6\sqrt{2})$ 3) $(\sqrt{3}, -\sqrt{6})$ 4) $(3\sqrt{6}, 6\sqrt{2})$
75. If $\bar{u}, \bar{v}, \bar{w}$ are unit vectors satisfying $2\bar{u} + 2\bar{v} + 3\bar{w} = \bar{0}$, then $|\bar{u} - \bar{v}|$ is equal to
- 1) $\frac{7}{4}$ 2) $\frac{\sqrt{5}}{2}$ 3) $\frac{\sqrt{7}}{2}$ 4) $\frac{5}{4}$
76. If A and B are different matrices satisfying $A^3 = B^3$ and $A^2B = B^2A$, then
- 1) $\det(A^2 + B^2)$ must be zero
2) $\det(A - B)$ must be zero
3) $\det(A^2 + B^2)$ as well as $\det(A - B)$ must be zero.
4) At least one of $\det(A^2 + B^2)$ or $\det(A - B)$ must be zero
77. Let S be the set of all $a \in \mathbb{R}$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}$, ($b > 1$) is acute. Then S is equal to
- 1) $\left(-\infty, -\frac{4}{3}\right)$ 2) Φ 3) $\left(-\frac{4}{3}, 0\right)$ 4) $\left(\frac{12}{7}, \infty\right)$
78. If $\vec{a} = 4\hat{i} + 6\hat{j}$ and $\vec{b} = 3\hat{j} + 4\hat{k}$, then the orthogonal projection of $\vec{a}$ along $\vec{b}$ is
- 1) $\frac{18}{10\sqrt{3}}(3\hat{j} + 4\hat{k})$
2) $\frac{18}{25}(3\hat{j} + 4\hat{k})$
3) $\frac{18}{\sqrt{13}}(3\hat{j} + 4\hat{k})$
4) $(3\hat{j} + 4\hat{k})$

79. Let A and B be two events such that $P(B|A) = \frac{2}{5}$, $P(A|B) = \frac{1}{7}$ and $P(A \cap B) = \frac{1}{9}$, Consider
 (S1) $P(A' \cup B) = \frac{5}{6}$, (S2) $P(A' \cap B') = \frac{1}{18}$. Then
 1) Both (S1) and (S2) are true 2) Both (S1) and (S2) are false
 3) Only (S1) is true 4) Only (S2) is true
80. Let x, y be real numbers satisfying $\sin x + \cos y = 1$ and $\sin y + \cos x = -1$. Then which of the following must be correct?
 1) $\sin(x+y) = 1$ 2) $\cos(x-y) = 0$
 3) $\cos(x+y) = 1$ 4) $\cos 2x = \cos 2y$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let the coefficients of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}} + \beta x\right)^4$, $(1 - 3\beta x)^2$ and $\left(1 - \frac{\beta}{2}x\right)^6$, $\beta > 0$, respectively form the first three terms of an A.P. If d is the common difference of this A.P., then $50 - \frac{2d}{\beta^2}$ is equal to _____
82. A class contains b boys and g girls. If the number of ways of selecting 3 boys and 2 girls from the class is 168, then $b + 3g$ is equal to _____
83. $\lim_{h \rightarrow 0} \frac{\ln(1+2h) - 2\ln(1+h)}{h^2} = K$, then the absolute value of K is _____
84. If $1 + (2 + {}^{49}C_1 + {}^{49}C_2 + \dots + {}^{49}C_{49})({}^{50}C_2 + {}^{50}C_4 + \dots + {}^{50}C_{50})$ is equal to $2^n \cdot m$, where m is odd, then $n + m$ is equal to _____
85. If A is a square matrix such that $A(\text{adj}A) = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$, then $\frac{|\text{adj}(\text{adj}A)|}{|\text{adj}A|}$ is equal to _____
86. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is _____

87. Let $S = \left[-\pi, \frac{\pi}{2}\right) - \left\{-\frac{\pi}{2}, -\frac{\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}\right\}$. Then the number of element in the set $A = \left\{\theta \in S : \tan \theta (1 + \sqrt{5} \tan(2\theta)) = \sqrt{5} - \tan(2\theta)\right\}$ is _____
88. Let $z = a + ib$, $b \neq 0$ be complex numbers satisfying $z^2 = \bar{z} \cdot 2^{1-|z|}$. Then the last value of $n \in \mathbb{N}$, such that $z^n = (z+1)^n$, is equal to _____
89. A bag contains 4 white and 6 black balls. Three balls are drawn at random from the bag. Let X be the number of white balls, among the drawn balls. If σ^2 is the variance of X , then $100\sigma^2$ is equal to _____
90. The value of the integral $\int_0^{\frac{\pi}{2}} 60 \frac{\sin(6x)}{\sin x} dx$ is equal to _____

